Wasi Omar Syed

CONTACT

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EDUCATION

Islamic University of Technology+01613346566 wasiomar21@gmail.com https://github.com/WasiOmar

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Structured

Programming

Systems: Database Management Systems I & II

Math Foundations: Discrete Mathematics, Linear Algebra, Differential CalculusRelevant CourseworkBachelor of Science in Software Engineering

CGPA 4.0 / 4.0 (4 semester)thBoardbazar, Gazipur

09/2024 - 09/2028

Programming: Python, C++, Java, JavaScript

Machine Learning & AI: PyTorch, Scikit-learn, YOLO, LangChain, LangGraph, CrewAI,

FastMCP

Data Science & Visualization: NumPy, Pandas, Matplotlib, Jupyter Notebook

Backend & Deployment: FastAPI, Node.js, Express.js , Next.js

Frontend & App Development: React, Next.js, Tailwind CSS, Flutter

Databases: MongoDB, PostgreSQL

DevOps & Cloud: AWS (basic services), Docker, Kubernetes (fundamentals), Terraform

(IaC)

Tools & Platforms: Git, GitHub, n8n

Mathematical & Statistical Foundations: Probability, Bayesian Inference, Hypothesis

Testing, Markov Models, Statistical InferenceTechnical Skills

CERTIFICATIONS

Harvardx CS50AI

Completed 12 AI projects using Python

Covered search algorithms, knowledge representation, optimization, machine

learning, neural networks, and NLP

Data Structures & Algorithms in C+ + –Apna College (2025)

Problem Solving & Competitive Programming

YouKnowWho Academy– Completed comprehensive training in algorithms, data

structures, and problem-solving techniques (2025).

Achievement

Project: PawFect Match – AI-powered pet adoption platform

Served as Project Lead, overseeing system design and implementationJuly 2025

IUTCS CodeSprint 2025 – Hackathon (Runners-up)Hands-on Machine Learning (from scratch implementations)

Implemented core ML algorithms from scratch using Python & NumPy (Linear/Logistic

Regression, Gradient Descent, PCA, Decision Trees, etc.)

Recreated key models without libraries to deeply understand mathematical foundations

and optimization techniques

Applied ML techniques on real datasets (Titanic, California Housing, Customer Churn,

etc.)

Compared classical ML models (e.g., Decision Trees vs Random Forest, Lasso vs Ridge)

Built full ML pipeline notebooks including EDA, preprocessing, training, and evaluationGitHub: github.com/WasiOmar/hands-on-mlDeep Learning Lab – CNNs, Dense Networks & Transformers

Implemented Transformer from scratch (Encoder + Decoder + Positional Embedding)

Built and trained Convolutional Neural Networks (CNNs) for image classification using

CIFAR-10 dataset

Compared traditional Neural Networks vs CNNs for image recognition performance (Mini

Project: Image Classification)

Implemented Dense Neural Network-based image classifier and evaluated on benchmark

dataset

Applied Transformer-based architecture for NLP task (IMDB sentiment analysis)

Gained practical understanding of deep learning architectures across vision and NLP

domainsGitHub: github.com/WasiOmar/Deep-Learning-Lab

FeatherDB | C++17 | Custom Relational Database Engine

Role: Parser & Query Engine Lead | https://github.com/SaifSarker11/FeatherDB

• Designed and implemented the full SQL parsing pipeline from scratch: tokenizer → AST →

query executor — no external libraries

• Built a recursive query executor supporting nested subqueries in both FROM and WHERE clauses

• Implemented in-memory ORDER BY sorting via Quicksort

• Enforced relational integrity at execution time: PRIMARY KEY, UNIQUE, and FOREIGN KEY constraints with RESTRICT delete policy